

Parts List for DCC Booster Rev B, DMFE 2020

Ref. Des.	Desc.	DK	Mfg.	
U1	Switching Reg. TO220-5	MC34167TVGOS-ND	MC34167TVG	
	Heat Sink	HS336-ND		
	Heat Sink	294-1082-ND		
U2	Switching Reg. 5v	102-5018-ND	P78E05-1000	
U3	74HC08 SOIC-14			
U4,U5	IR2101S 8SOIC	IR2101SPBFCT-ND	IR2101STRPBF	
U6, U11	74AHC1G04GV SC74A	1727-4199-1-ND	74AHC1G04GV.125	
U7	6N137 (HCPL-0601)	516-1774-1-ND	HCPL-0601-500E	
U8	PIC16F1847-IP DIP-18			
U9	74AHC1G86GV.125	1727-4201-1-ND	74AHC1G86GV.125	5TSOP
U10	LM358AM SOIC-8			
U12,U13	LMC555 SCIC-8	LMC555CMX/NOPBCT-ND	LMC555CMX/NOPB	
L1	180uH 5A AIRD-03-181	535-11334-ND	AIRD-03-181K	
Q1,Q2,Q3,Q4	MOSFET N-CH 60V 20A	NTD5867NLT4GOSCT-ND	NTD5867NLT4G	DPAK
D1,D2,D3,D4,D6, D7,D10,D11,D12,D13 D5,D27	Diode Schottky (10) Diode 1A SMA	SBR545D1-13DICT-ND	SBR545D1-13	DPAK
D8,D9,D14, D21,D22,D23,D24 D25,D26	1N4148W Zener 15V SOD-123	SOD-123 BZT52C15-FDICT-ND	BZT52C15-7-F	
D15,D16,D17,D18,D19, D20	LED Green x3 (2)	754-1289-ND	WP934SA/3GD	
FB1,FB2	100Ω 1812	732-3431-1-ND	74279226101	
TVS1	TVS 17V	SMBJ17CA-FDICT-ND	SMBJ17CA-13-F	
C1,C5,C7,C13, C14,C15,C16,C18,C19, C20,C21,C22	0.1uF 50v Chip 0805 (13)			
C2	10uF 50v Chip	587-5961-1-ND	UMK325BJ106KMHP	SMD 1210
C3	22uF 25v Chip	1276-3373-1-ND	CL32A226MAJNNNE	SMD 1210
C4	470uF 63v Electrolytic	493-1949-ND	UPW1J471MHD3	
C6	4700uF 25v Electrolytic	1189-1732-ND	25PK4700MEFC16X25	
C8,C9	1.0uF 63V poly	399-5447-1-ND	R82DC4100DQ60J	
C10,C11	10uF 25V Tant	478-5257-1-ND	TAJB106K025RNJ	
C12	100pF 25V 0805 (Optional)	399-1122-1-ND	C0805C101J5GACTU	
C17	1.0uF 25V 0805			
C23,C24	220uF 50v Electrolytic	PCE3921CT-ND	EEE-1HA221P	SMT 10mm
C25,C26	0.01uF 50V 0805	1276-1078-1-ND	CL21B103KBCNNNC	
SW1	Switch DIP 5	GH7138-ND	76PSB05T	
J1	Header 4 pin	ED2830-ND	OSTOQ045451	
J1.1	Plug 4 pin	ED2910-ND	OSTTJ045153	
J2,J3	RJ-9 Jack	AE10381-ND	A-2004-3-4-LP-N-R	
R1	68K 0805			
R2	3.3K 0805			
R3,R17	6.8K 0805 (2)			
R4,R5,R6,R7,R39,R40 R8,R14,R16,R29,R30, R33,R35,R36	10Ω 0805 (6) 10K 0805 (8)			
R9,R18,R19,R20,R21, R31	1.0K 0805 (6)			
R10,R11	470Ω 0805 (2)			
R12,R13	4.7K 0.75W 2010 (2)	RMCF2010JT4K70CT-ND	RMCF2010JT4K70	
R15,R34	0.1Ω 1W 2512 (2)	CRA2512-FZ-R100ELFCT-ND	CRA2512-FZ-R100ELF	
R22	15K 0805			
R23	4.7K 0805			
R24,R25,R26,R27,R28	47K 0805 (5)			
R32	100K 0805			
R37,R38	100Ω 0805			

Notes:

Voltage: $V_{Out} = (R3/R2 + 1) * 5.05V$ $6.8K/3.3K = 15.32V$ @ TP+15V = 14.5 VAC no load = 13.55VAC @ 0.75A
 $6.8K/3.0K = 16.4V$

Changes for Rev ?: